

What does the edge cost? Capital-adjusted market making under a public portfolio-margin engine

Gregor Albiez^a

^aUniversity of Applied Sciences and Arts Northwestern Switzerland (FHNW), Switzerland

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ABSTRACT

A market maker quotes against a budget of capital, not contracts, yet its edge is reported per contract. On Derive, a crypto options exchange, the margin rules are public smart contracts, and its off-chain engine quotes margin for any book. Capital is what its three margin managers require for positions, not account balances; it is measured on one contract per option fill of a companion paper and on the actual books of dominant maker subaccounts. Four hypotheses were registered before any registered capital figure existed. Contrary to H1, capital does not reorder the moneyness-by-tenor map of the edge (rank correlation 0.903); post hoc, much of that correlation comes from the sign of the edge. In the books of the four subaccounts under the second-generation portfolio manager (PM2), trading only ETH and HYPE, the median fill binds 3.5 per cent of its stand-alone capital (H2 not rejected). On the opening books of nine dominant subaccounts, standard margin needs 4.75 times the PM2 capital (H3 not rejected), mostly for books too large for a standard-margin subaccount. Contrary to H4, cheaper capital did not detectably narrow the half spread; the test has little power. All numbers use pilot data to 17 September 2026.

1. Introduction

A market maker on an options venue does not run out of contracts. It runs out of capital. Funding constraints act on the deposit that backs the margin of its positions (Brunnermeier and Pedersen, 2009), and margin-based pricing turns that deposit into a required return (Gârleanu and Pedersen, 2011). The literature on market making nonetheless counts in contracts or their value. Inventory models penalise the position (Ho and Stoll, 1981; Avellaneda and Stoikov, 2008) or bound it in units (Guéant et al., 2013). Option market making prices the risk of the book (Stoikov and Sağlam, 2009; Gârleanu et al., 2009), the spread compensates for risks the maker cannot hedge (Jameson and Wilhelm, 1992), and inventory risk moves option prices (Muraviev, 2016). Even the illiquidity premium in options is measured against the option price (Christoffersen et al., 2018). Capital enters option prices through the maker's wealth (Fournier and Jacobs, 2020) or aggregate intermediary constraints (Chen et al., 2019). What a unit of edge costs in capital remains open: outsiders can rarely combine an exact margin rule with the books makers actually hold.

Derive, a crypto options exchange, makes both observable. It margins every subaccount under one of three managers: a strategy-based standard manager (SM), a legacy portfolio manager (legacy PM) and a second-generation portfolio manager (PM2). Each is a deterministic function of the positions, the price feeds and risk parameters stored on chain, and its smart contracts can be queried at any past block. A public endpoint evaluates any hypothetical book with the venue's off-chain trading engine. Makers' holdings are on chain too, so their actual books can be priced. Capital is measured on positions under the venue's rules, not on

account balances, which mix what a book needs with what a trader deposits.

The question extends Albiez (2026), which maps the maker's net edge over moneyness and tenor. Here the same fills are priced in capital: how much capital does the edge bind, and does the map change when the edge is measured per unit of capital? Option strategies lose much of their appeal once margin is charged (Santa-Clara and Saretto, 2009); this paper takes the maker's side and uses the rule the venue actually applies.

The paper makes four contributions: a capital map over the moneyness and tenor grid of Albiez (2026); the marginal cost of a fill in the actual book of a dominant subaccount; the value of netting on the same books under standard margin and PM2, in the tradition that compares strategy-based and portfolio margining on identical positions (Kupiec and White, 1996); and a price of capital. Makers' inventories and revenues move liquidity (Comerton-Forde et al., 2010), and margin changes move swaption prices (Ahn, 2025); here, parameter changes form a quasi-experiment: on-chain events that change the stand-alone capital of every cell by a known amount.

Four hypotheses were registered before any capital figure of the registered analysis existed: that edge per notional and edge per unit of PM2 capital rank the cells differently (H1); that a fill in the book of a dominant subaccount under PM2 binds less than half its stand-alone capital per contract (H2); that PM2 saves more than half the capital standard margin requires for the same books (H3); and that the half spread falls when a parameter change makes capital cheaper (H4). Addenda 1 to 7 of the pre-registration are described before the appendices.

H1 and H4 are rejected; H2 and H3 are not. The capital denominator does not reorder the map in the registered sense: the rank correlation is 0.903, with an interval from

✉ gregor.albiez@students.fhnw.ch (G. Albiez)
ORCID(s): 0009-0006-6036-4932 (G. Albiez)

BTC · 17 Sep 2026, 08:00 UTC · 15 live expiries · PM2 spot grid $\pm 14\%$ since 20 Aug 2026 · on-chain engine

a PM2: ATM 30 d short = 11.8 % of forward

b SM: ATM 30 d short = 14.1 % of forward

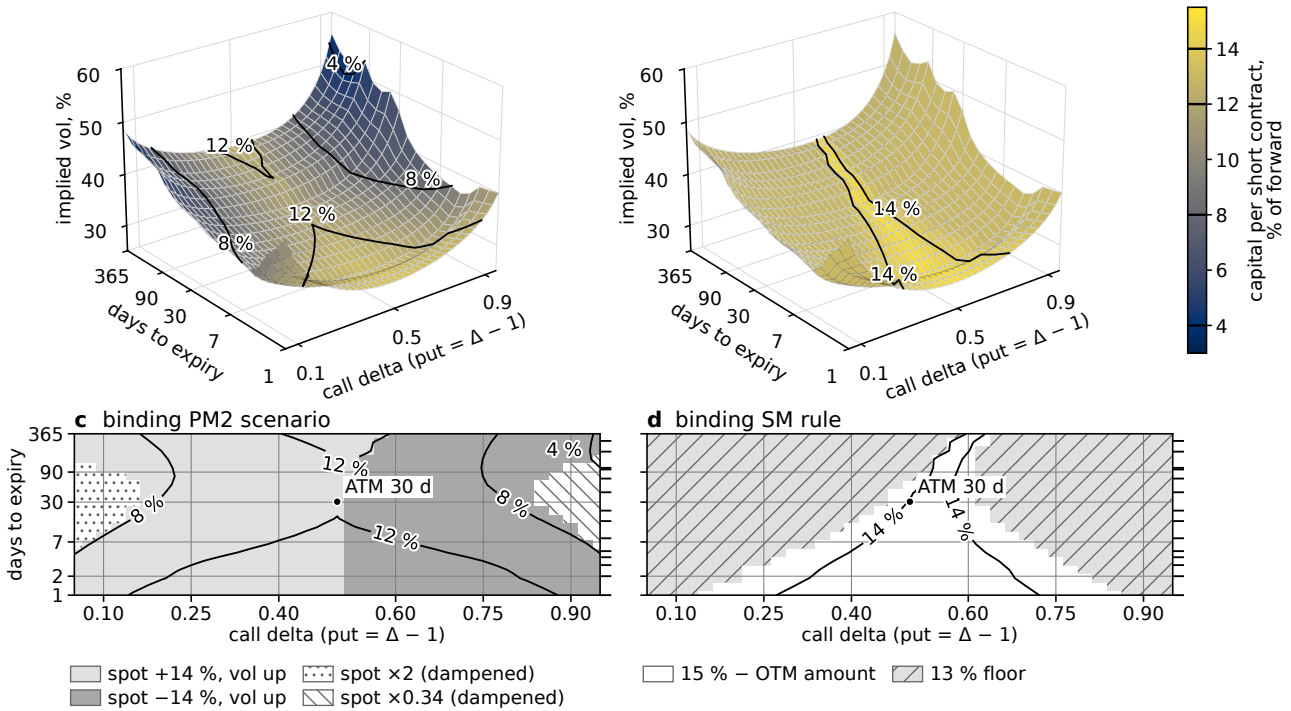


Figure 1: The surface as the engine sees it. BTC implied volatility over call delta and tenor at 08:00 UTC on 17 September 2026, built from the on-chain feeds, with the capital that one short contract binds under PM2 (panel a) and under standard margin (panel b) as colour on a common scale in per cent of the forward. Height is volatility, colour is capital, and black lines join points of equal capital. ATM marks the at-the-money node at 30 days to expiry. Panels c and d name the rule that sets the capital at each point: the worst scenario of the PM2 grid, and the branch of the standard margin formula, a share of spot less the out-of-the-money (OTM) amount or a floor. Their grid lines are the tenor and absolute delta bucket edges of Figures 3 and 4, and the ticks on their right edge mark the listed expiries. For puts the delta edges are mirrored: the lines at 0.60, 0.75 and 0.90 are the put edges of 40, 25 and 10 per cent. The surface holds out-of-the-money options only, so the buckets from 60 per cent absolute delta up have no counterpart here. Standard margin is set in per cent of spot and shown in per cent of the forward. Capital follows the smart contracts on chain; the venue's off-chain engine discounts PM2 at a flat two per cent.

0.881 to 0.907 against a threshold of 0.5 (H1). Post hoc, much of this correlation comes from the sign of the edge: among profitable cells it is 0.634 (0.572 to 0.702). In the books of the four subaccounts under PM2, which trade ETH and HYPE only, the median fill binds 3.5 per cent of its stand-alone capital (H2); maker buys there mostly release capital and sells mostly bind it. Standard margin needs 4.75 times the PM2 capital for the opening books of nine dominant subaccounts (H3), most of these books larger than a standard-margin subaccount accepts. I find no evidence that the half spread fell when capital got cheaper (H4): the estimate has the wrong sign and a one-sided p of 0.62, but the test cannot exclude a sizeable narrowing.

2. The engine and what it returns

This section derives the capital measure from the margin rules. The standard manager is strategy based. Each option is margined in isolation, each expiry takes the smaller of its summed option margins and its maximum loss, and expiries add up. A long option earns no credit of its own,

so under SM a long book's capital is its full premium. The legacy manager and PM2 are scenario based, in the spirit of Standard Portfolio Analysis of Risk (Kupiec, 1994): the book is revalued under spot and volatility shocks, and the requirement is the worst loss, scaled by a margin factor, plus contingencies (panel c of Figures 1 and 2). Both haircut shocked expiry values with a static discount. PM2 adds skew scenarios, applies the haircut to positive and negative expiry values alike, and discounts at a live rate feed; the legacy manager's rate stays at zero. Both net within one underlying and never across, and the scope of netting decides what it saves (Duffie and Zhu, 2011; Cont and Kokholm, 2014). Probability-based margin goes back to Figlewski (1984), and the worst-loss core of scenario margin is a coherent risk measure (Artzner et al., 1999). The managers' smart contracts are public (Derive, 2026b), with parameters on chain as in the lending protocols of Qin et al. (2021); BitMEX instead margins each position at a percentage of its notional (Soska et al., 2021).

The public endpoint `get_margin` (Derive, 2026a) does not return a requirement. For positions q and USDC collateral C

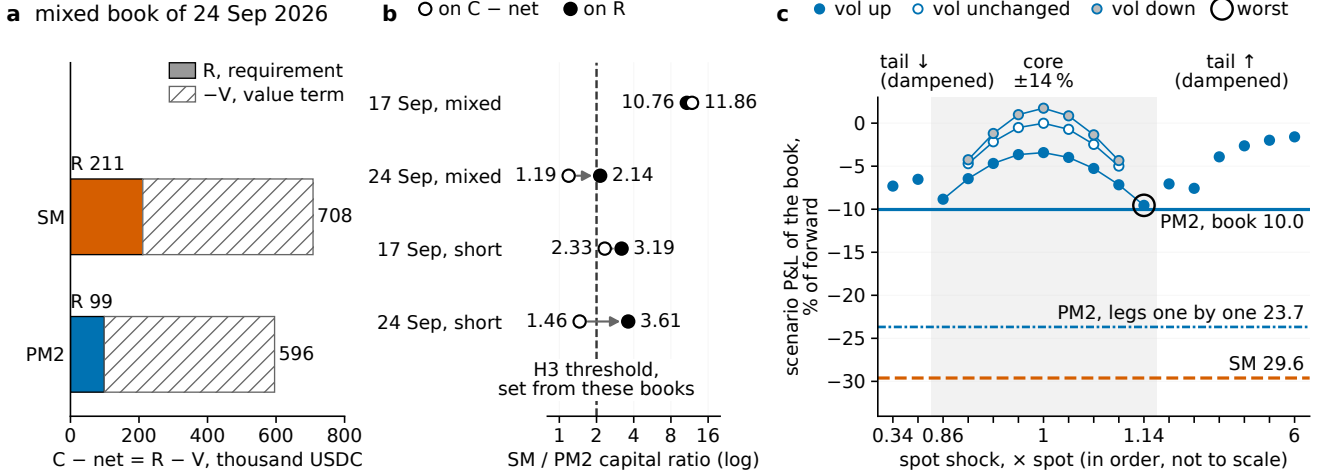


Figure 2: What `get_margin` returns, and how PM2 prices a book. Panel a splits $C - \text{net}$ into the requirement R and the value term $-V$ for the mixed probe book of 24 September 2026 (block 45 110 142); $-V$ is almost the same under both managers and dwarfs R . Panel b sets the ratio of standard to PM2 margin read as $C - \text{net}$ (open) against the ratio on R (filled) for the four probe books at their historical blocks. A short book is short in every option it holds, a mixed book long in some and short in others. The dashed line is the H3 threshold of two, which was set from these books before any maker book was measured. On the mixed book of 17 September, V is positive, so the ratio falls from 11.86 to 10.76. Panel c is the scenario profit and loss of a short BTC straddle struck at the forward on 17 September 2026, on the listed expiry nearest to 30 days (22 days), in per cent of the forward, over the spot shocks in order, not to scale. PM2 charges the worst scenario of the whole book plus contingencies (solid line), which is less than the sum of the legs margined one by one (dash dot); standard margin is dashed.

it returns, for initial and for maintenance margin X ,

$$\text{net}_X(q; C) = C + V(q) - R_X(q), \quad (1)$$

with V the value of the positions in the manager's own valuation and R_X the requirement. The venue presumably admits a book whose initial net margin is non-negative; the chain enforces a weaker check, so binding initial margin is an assumption. The fields `pre` and `post` evaluate the book as given and after simulated changes at one price state within one request, which makes a marginal cost measurable.

Treating $C - \text{net}$, which equals $R - V$, as the requirement would mix the value V into the margin. Before the registration, this gave ratios of standard to PM2 margin of 11.86 and 1.19 for the mixed probe books of 17 September and 24 September 2026; on R the ratios are 10.76 and 2.14 (Figure 2). The second book's PM2 value of $-496\,859$ USDC adds almost the same amount to numerator and denominator, pulling the ratio towards one. Netting factors are properties of books, so this paper measures them on real ones.

The quantity used throughout is the capital needed to open a book q at traded prices p and meet initial margin,

$$K_p(q) = \sum_{i \in \text{options}} p_i q_i - \text{net}_{\text{IM}}(q; 0). \quad (2)$$

Collateral enters net margin one for one, so K_p is the deposit at which the book is exactly admissible once the premium has changed hands. The measure needs no valuation of its own: it differs from the requirement R_{IM} by $\sum p_i q_i - V$, a valuation convention, not capital. For a trade Δq at prices

p , one request with a collateral change of $-p \Delta q$ gives $\Delta K = -(\text{post} - \text{pre})$. Figure 1 maps K_p of single contracts on the BTC surface.

Two engines implement the managers. The off-chain engine behind the endpoint, on which the venue trades, prices at fresh ticker marks and discounts PM2 at an undocumented flat rate; a box spread in each of the fourteen listed BTC expiries on 24 September 2026 puts that rate at 2.0000 per cent. The on-chain engine settles and liquidates, reads feeds that lag the ticker, and discounts with the rate feed, which ranged from 3.64 to 3.82 per cent across the same expiries. Only the on-chain engine can be rebuilt for past blocks, so past fills are priced at their block with offline replicas of its smart contracts. On 195 single contracts on 25 September 2026, the two engines differed in PM2 capital by a median of 0.08 per cent and at most 2.4 per cent, at the longest tenor; whole books, probed at random, showed larger gaps.

3. Data and measurement

The fills are those of Albiez (2026): all 603 940 BTC, ETH and HYPE option fills from 11 January 2024 to the pilot cut of 17 September 2026. Each carries the maker's side, price, size and net edge in USDC per contract: the markout against the mark thirty minutes after the fill, less fee net of rebate and a hedging cost. Fee and rebate, reported per fill, are divided by the fill's size (Addendum 2). On a multi-leg request for quote (RFQ) the venue books the package fee on one leg, where it stays (Addendum 6). The companion paper was corrected on 25 September 2026 and uses the same

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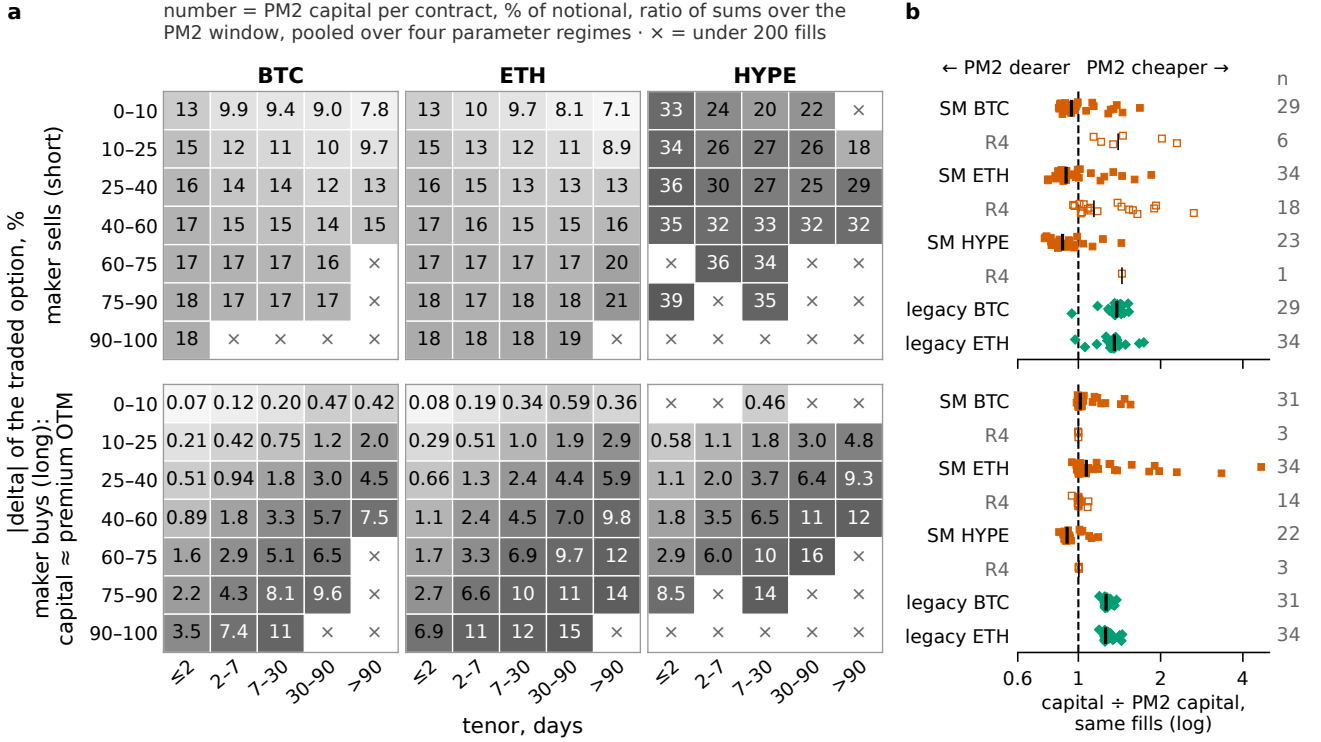


Figure 3: What one contract costs. Panel a is PM2 capital per contract in per cent of notional over the absolute delta of the traded option and tenor, for maker sells (upper row) and maker buys (lower row), as a ratio of sums over the fills of the PM2 window, which the parameter changes of 23 January, 24 May and 20 August 2026 split into four regimes, R1 to R4. Each row is shaded on one logarithmic grey scale, and an empty cross marks a cell with fewer than 200 fills. Panel b gives, for every occupied cell, the capital of the same fills under standard margin (squares) and under the legacy manager (diamonds) divided by PM2 capital, with the median cell as a bar and the number of cells under n ; the hollow squares of the rows R4 use only the fills after the parameter change of 20 August 2026. For a maker buy, standard margin charges the premium, and PM2 about the premium out of the money (OTM) and less in the money.

unit per contract. SM covers the whole sample (HYPE from 11 November 2025; six earlier HYPE fills have no manager), the legacy manager BTC and ETH throughout. The PM2 window is the span in which PM2 covers an underlying: BTC and ETH from 12 June 2025 23:00 UTC, HYPE from 11 November 2025. Only H4 uses events before it.

A fill's market state is rebuilt from the parameter timelines and the last feed values pushed by its block. The fill's stand-alone capital is that of a book holding only the fill, at its price, under every manager available at the block: $q = +1$ for a maker buy and $q = -1$ for a sell (Figure 3). A cell combines an underlying, a maker side and one absolute delta and one tenor bucket of Albiez (2026); a cell with at least 200 fills is occupied. Its edge per unit of capital is a ratio of sums in basis points,

$$e_c^K = 10^4 \frac{\sum_{i \in c} NE_i a_i}{\sum_{i \in c} K_i a_i}, \quad (3)$$

with NE the net edge and a the size; its edge per unit of notional has the index price in place of K .

The maker books are those of the ten subaccounts with the most maker fills, M1 to M10 by rank. Their opening books are read on chain, with their manager, at the first block of each UTC day. Some share a wallet: M3 with

M5, M4 with M10, and M2, the one SM subaccount, with M1, M6 and M8. The book before a fill is the opening book plus the subaccount's balance changes up to the fill's transaction (Addendum 1). The marginal cost of a fill is $\Delta K = K(\text{after}) - K(\text{before})$ per contract under the subaccount's manager, relative to its stand-alone capital. The netting value of a maker-day is the SM capital of its opening book over its PM2 capital, each summed over underlyings, with positions at the companion paper's mark.

For H4 each parameter change j becomes a dose $d_{c,j}$: the mean of $\log(K^+/K^-)$ over the fills of cell c in the fourteen days before the event, where K^+ and K^- are the capital of a fill after and before it in the same market state. Doses are shown in log per cent, 100 times the log change; events whose largest absolute dose is below one log per cent are dropped. Over fourteen days on either side of the event, excluding its day, the half spread (the markout at the fill) in basis points of the index follows

$$y_i = \alpha_{c,j} + \gamma_{t,u} + \beta \text{post}_i d_{c,j} + \varepsilon_i, \quad (4)$$

with post_i marking fills after the event, t the UTC day and u the underlying; cells need 20 fills on each side.

Intervals are 90 per cent percentile intervals of a cluster bootstrap over UTC days with 9999 draws (Cameron et al.,

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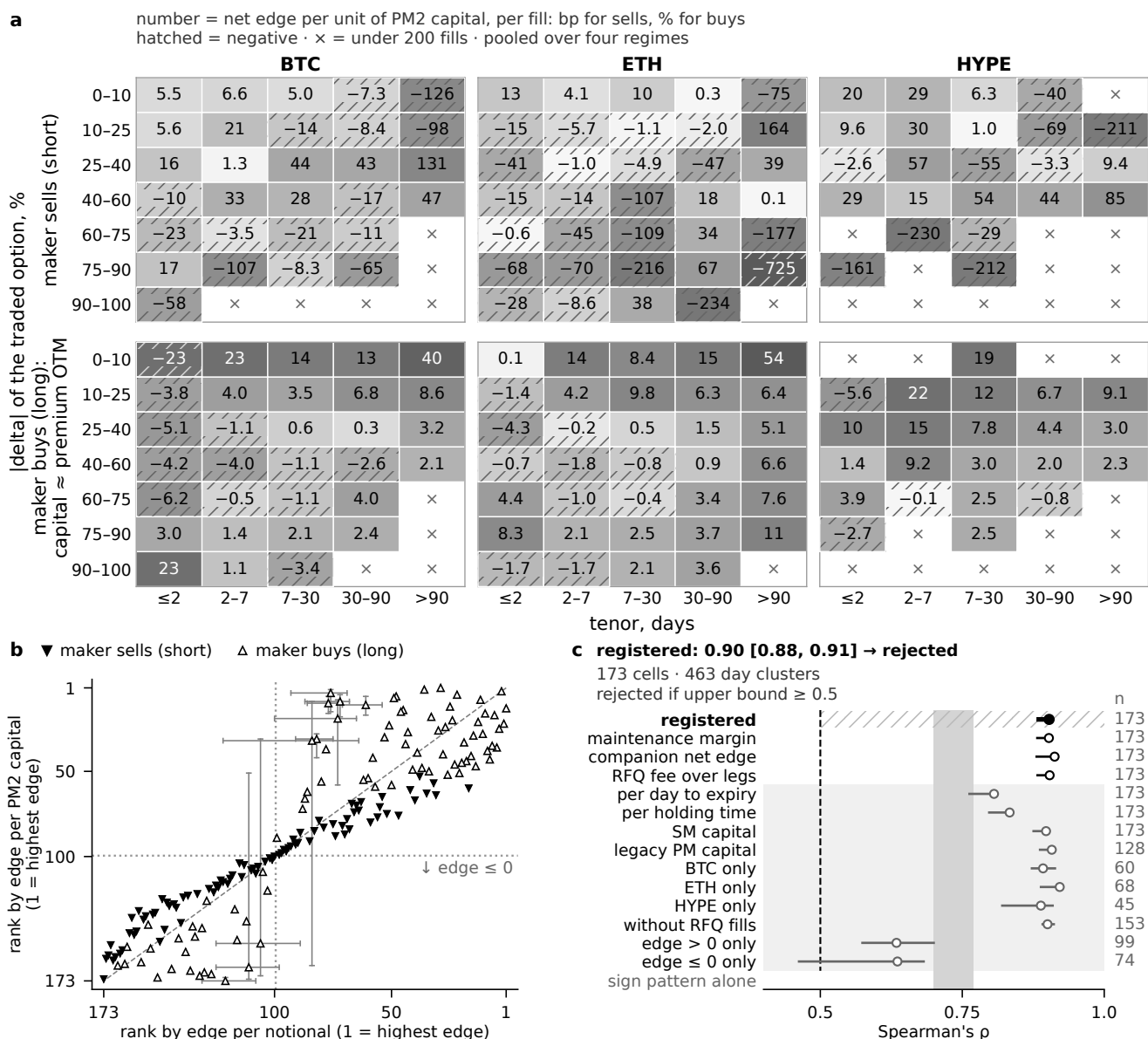


Figure 4: The map in two denominators (H1). Panel a is net edge per unit of PM2 capital, per fill, for every cell of the PM2 window by underlying and maker side, in basis points for maker sells and in per cent for maker buys; negative cells are hatched and an empty cross marks fewer than 200 fills. For a maker buy the denominator is about the premium when the option is out of the money and less in the money. Darker grey marks a larger absolute value on a log scale, each row on its own. The map per notional is not drawn; its ranks enter panel b. Panel b ranks the 173 occupied cells by edge per notional and by edge per PM2 capital; because capital is positive in every cell, the 99 cells with positive edge come first in both rankings, and the dotted lines mark that boundary. Crosses give the 90 per cent rank intervals of the ten largest moves. Panel c is the registered test: Spearman's ρ with its 90 per cent day-cluster interval (bar, also printed above the panel), set against the threshold of 0.5. The hatched stretch of the registered row, right of the threshold, is the rejection region: the rule rejects H1 if the interval reaches into it. Sensitivities follow and then, on grey, exploratory rows; n is the number of cells in a row, and in the rows by the sign of the edge the cells are chosen again in every replicate. The grey band is the ρ of the sign pattern alone, with ranks shuffled within each sign group.

2008). H4 uses a wild cluster bootstrap with Rademacher weights and restricted residuals (MacKinnon and Webb, 2017; Roodman et al., 2019). It uses 100 placebo dates, each at least 28 days from every change of the underlying's legacy-manager parameters and default PM2 parameter set (subaccounts may carry their own; Addendum 4). Replicas

were validated against the chain's evaluation (eth_call) before any capital entered a test (Appendix A). Analyses not named in the registration are exploratory, those added after the first results post hoc.

registered: 0.035 [0.031, 0.039] → not rejected

4 PM2 subaccounts, ETH and HYPE only
19 999 of 20 000 sampled fills · 372 day clusters

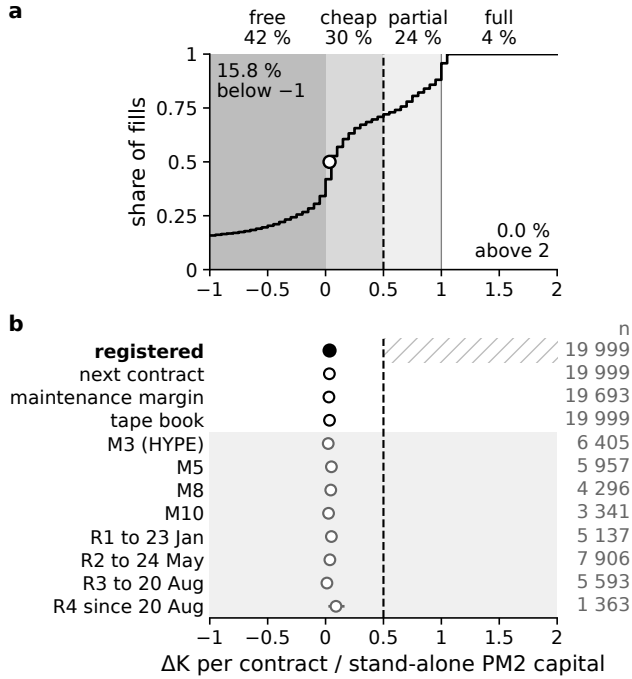


Figure 5: A fill in a dominant maker's book (H2). Panel a is the cumulative distribution of the marginal capital of a fill per contract over its stand-alone PM2 capital, $\Delta K/K_{\text{single}}$, for the 19 999 tested fills of the four PM2 subaccounts, which trade ETH and HYPE only; the numbers above the bands are the shares of fills in each band, and the shares beyond the axis are given at both ends. The dashed line is the registered threshold of one half, and the circle at height one half is the median; its 90 per cent day-cluster interval is narrower than the circle and printed at the top. Panel b repeats the registered row above the sensitivities and, on grey, exploratory rows per subaccount and per parameter regime. The hatched stretch of the registered row, right of the threshold, is the rejection region: the rule rejects H2 if the interval reaches into it. A maker who starts from an empty book pays the stand-alone capital of Figure 3.

4. Results

4.1. The capital denominator and the map (H1)

Under PM2 a single maker sell binds between 7.8 and 17.7 per cent of notional in BTC, between 7.1 and 21.4 in ETH and between 17.5 and 38.7 in HYPE (Figure 3); a maker buy binds about its premium out of the money and less in the money. Pooled over the window, PM2 is not the cheap manager for one short contract: the ratio of standard margin to PM2 capital in the median sell cell is 0.943 in BTC, 0.902 in ETH and 0.875 in HYPE. After the parameter change of 20 August 2026 the ratio is 1.398 for BTC and 1.140 for ETH, on the 6 and 18 sell cells still occupied. The legacy manager needs more, with medians of 1.382 and 1.355 for BTC and ETH sells.

In the PM2 window 173 cells are occupied, out of 210 with any fill (Figure 4). Across them, Spearman's rank correlation ρ between edge per notional and edge per unit of

registered: 4.75 [4.66, 4.82] → not rejected

9 subaccounts · 1 943 maker-days
462 day clusters (UTC days, not subaccounts)

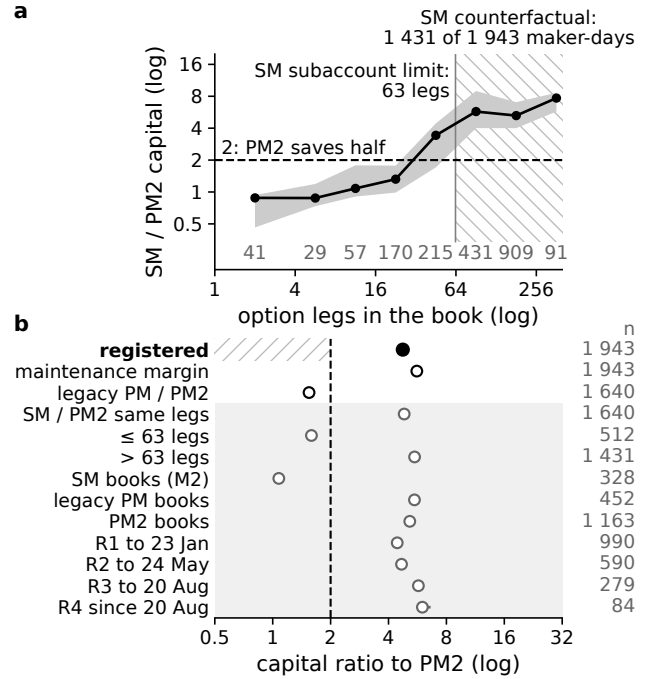


Figure 6: What netting is worth (H3). Panel a is $K_{\text{SM}}/K_{\text{PM2}}$ for the opening books of 1 943 maker-days against the number of option legs, as the median per bin with the interquartile band; the grey numbers above the axis are the maker-days per bin. Beyond 63 legs, the most the endpoint accepted for a standard-margin subaccount in probes of September 2026, K_{SM} is counterfactual (hatched); this is the case on 1 431 maker-days. The dashed line is the registered threshold of two. Panel b is the registered median with its 90 per cent interval, the sensitivities including the legacy manager on BTC and ETH legs, and, on grey, exploratory rows: SM against PM2 on the same BTC and ETH legs, by book size, by the manager of the subaccount and by parameter regime. The hatched stretch of the registered row, left of the threshold, is the rejection region: the rule rejects H3 if the interval reaches into it.

PM2 capital is 0.903 (90 per cent interval 0.881 to 0.907); the upper bound exceeds 0.5, so the registered rule rejects H1.

The rejection says less than its margin suggests (post hoc, Appendix B): capital is positive in every cell, so a cell keeps the sign of its edge, and the 99 profitable cells rank first in both maps. Among them alone ρ is 0.634 (0.572 to 0.702), and among profitable buys 0.243 (0.207 to 0.446).

Pooled, the net edge per fill is 3.38 basis points of notional and 39.9 of PM2 capital, not a return per unit of time. The median cell moves by 14 ranks. The largest moves are maker buys far out of the money, whose small capital turns a thin edge per notional into a large edge per unit of capital. BTC buys in the lowest delta bucket with two to seven days to expiry rise from rank 76 to 4; the same buys with at most two days fall from rank 121 to 173, the last. Maker sells, which bind scenario capital, move less, and

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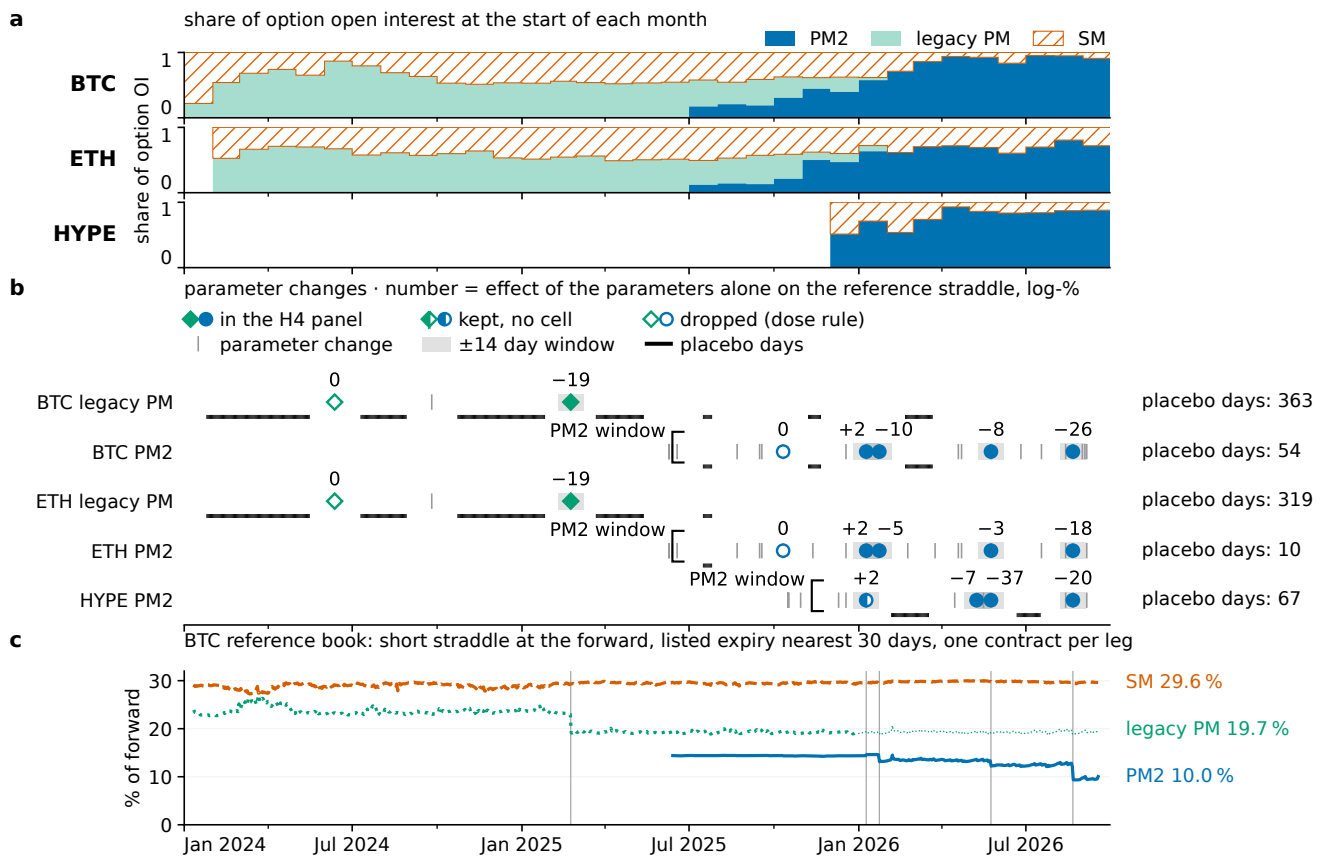


Figure 7: The engine over time. Panel a is the share of option open interest per manager and underlying at the start of each month. Panel b marks every parameter change of the legacy manager and of PM2 as a grey tick and the registered H4 events as symbols: filled when they enter the H4 panel, half filled when kept without a cell of 20 fills on each side, and hollow when dropped by the one log per cent dose rule. The number is the change that the parameters alone make to the capital of the reference straddle of the underlying, in log per cent. Grey bars are the windows of 14 days on either side of each kept event; the windows of January and of May 2026 overlap, so 6527 fills enter two events. Black dashes below each line are the days from which placebo dates may be drawn. Panel c is the capital of the BTC reference book, a short straddle struck at the forward on the listed expiry nearest to 30 days, one contract per leg, in per cent of the forward; the small saw teeth come from rolling between expiries of 21 and 36 days, and the legacy line is thin in months in which the legacy manager held less than 5 per cent of BTC open interest. The grey vertical lines in panel c mark the kept BTC events of panel b. Parameter changes of standard margin left the reference book unchanged.

none ranks higher than 53rd per unit of capital. At the top per unit of capital, the edge comes mostly from RFQ legs and is close to the gap between fill price and the stochastic volatility inspired (SVI) mark, which is extrapolated in the far wing (post hoc). The 20 fills with non-positive PM2 capital stay in the sums (Addendum 4).

4.2. A fill in a maker's book (H2)

Of the ten dominant subaccounts, four were margined under PM2 at the start of a day in the window (Addendum 1); they trade ETH and HYPE only. The registered random sample draws 20000 of their 100995 fills. The registered exclusion of one fill with non-positive stand-alone capital leaves 19999 fills on 372 days. Figure 5 shows the distribution of $\Delta K/K_{\text{single}}$, the marginal capital of the whole fill per contract. The median ratio is 0.0345, with a 90 per cent interval from 0.0307 to 0.0386, far below the registered threshold of one half, and 42.0 per cent of the fills leave the

book's capital unchanged or reduce it. H2 is not rejected: in these books the median fill costs 3.5 per cent of what it would cost alone.

The median hides a wide spread and mixes the sides: maker buys mostly release capital and maker sells mostly bind it. About a quarter of the fills release more than 0.28 of their stand-alone capital; a quarter bind more than 0.63 of it. From the 95th percentile on, a fill binds about its full stand-alone capital and almost never more. Fills that offset the loss in the book's worst scenario release capital; fills that add to it cost about what they cost alone. Per subaccount (exploratory) the median runs from 0.0230 for the single HYPE subaccount to 0.0505.

4.3. What netting is worth (H3)

Figure 6 reports the capital of the same opening books under SM and PM2. It covers 1943 maker-days in the PM2

window with at least one option position, from nine subaccounts: four under PM2, four under the legacy manager, for which both capitals are counterfactual, and one under SM; the tenth made no maker fill in the window. No book has non-positive PM2 capital. The median of K_{SM}/K_{PM2} is 4.747, with a 90 per cent interval from 4.662 to 4.824, and eight of the nine subaccount medians exceed two. H3 is not rejected; its registered threshold of two lies at the lower end of the probe-book ratios of Section 2 (2.14 to 10.76 on R). On BTC and ETH legs the legacy manager needs 1.545 times the PM2 capital (a comparison without a threshold), and standard margin 3.245 times the legacy capital.

The factor comes from large books and from the subaccounts that hold them. On 1 431 maker-days (73.6 per cent) the book holds more than 63 option legs, the most an SM subaccount can hold, so its SM capital is counterfactual (Addendum 4). On those days the median ratio is 5.465. On the 512 days with at most 63 legs it is 1.590 (exploratory, interval 1.532 to 1.662), a value the registered rule would reject. But 328 of those days belong to M2, the one SM subaccount. Its books net little (1.076), while the portfolio-margin subaccounts of its wallet hold large books. Without M2, the small books give 3.780 (post hoc); books of 32 to 63 legs, M2 included, give 3.43 (Figure 6, panel a). Whether a large book could be split across standard-margin subaccounts, as operators split books across managers, is not examined.

4.4. The price of capital (H4)

Figure 7 places the parameter changes next to the capital of a fixed reference book, a short straddle at the forward of each underlying (BTC in panel c). The registered set holds 18 events once same-day changes are merged, without changes that touch only collateral parameters or the number of expiries (Addendum 4); 14 pass the dose filter, and 13 have cells with at least 20 fills on each side, 475 cell-event pairs: the H4 panel. The legacy-manager changes of 22 February 2025 for BTC and ETH, before the PM2 window, supply 75 of them. Of the 14 kept events, 11 made the reference book cheaper, by between 3.4 and 37.4 log per cent across underlyings and between 7.8 and 26.0 for BTC. The largest BTC step, on 20 August 2026, cut the PM2 capital of the straddle by 26.0 log per cent (22.9 per cent). Only the events of 8 January 2026 raised it, by about 1.6 log per cent.

With its fixed effects, β compares, within the same event, cells whose capital changed more with cells whose capital changed less, mostly sells against buys (Figure 8, panel a). It assumes that such cells would have moved in parallel without the change and that makers did not anticipate it; the venue chose the changes itself. The dose is the change in stand-alone capital, while in a dominant book a fill binds far less (H2). The half spread is measured on fills, not quotes. At the kept events, the changed manager held between 54.5 and 94.9 per cent of the open interest.

The estimate of β is -4.60 basis points of the index per unit of log capital, with a one-sided wild cluster bootstrap p of 0.6224, and 64 per cent of the 100 placebo estimates are

at least as large (Figure 8). H4 is rejected unless the estimate is positive with a one-sided p of at most 0.05 and lies above the 95th placebo percentile, here 23.41. The estimate fails both criteria, and H4 is rejected. The sign is the opposite of the hypothesis, and with a cluster standard error of 13.09 the estimate cannot be told apart from zero.

The test has little power. Its standard error holds the event dates fixed, and at the 100 placebo dates the t statistic has a standard deviation of 1.82 instead of one (post hoc). Calibrated on those placebo statistics, the 90 per cent range for β runs from -41.6 to 29.7 . For capital ten per cent cheaper, the change in the half spread ranges from -3.13 to $+4.39$ basis points of the index, against a panel mean of 7.93: a narrowing by up to 39 per cent is not excluded. Only 141 day clusters carry the test, and no variant in Appendix B meets the registered rule.

4.5. Sensitivities

No sensitivity named in the pre-registration or its addenda changes a verdict. Under maintenance margin, ρ is 0.903 and the netting ratio 5.614. The median marginal cost is then 0.0291 on 19 693 fills with each subaccount's own PM2 parameter set (some carry a lower maintenance factor) and 0.0278 with the default set; using one set for both capitals of a ratio was decided after the first results. With the net edge as defined in the companion paper's first version, ρ is 0.913; with the fee of an RFQ package spread over its legs, it is 0.904. With the book rebuilt from the trade tape, the median marginal cost is 0.0340, and for the next single contract, which the registered wording also covers, 0.0329; Addendum 4 chose the whole fill once both measures existed, before any median was computed.

5. Discussion

For a maker or a quoting bot on this venue the results have practical consequences. First, capital leaves the order of the map largely intact (H1), but from an empty or small book the side matters more than the cell. A long option far out of the money binds about its premium, and a short one binds scenario capital. The same edge per notional is therefore a large edge per unit of capital when the maker buys a cheap wing and a small one when it sells. The top cells per unit of capital owe their edge mostly to RFQ legs and to the SVI mark extrapolated in the far wing (post hoc). Buys mostly release capital in the dominant PM2 books, where marginal capital orders the sides differently from stand-alone capital.

Second, the book, not the contract, decides the cost of a quote. In four large PM2 books the median fill binds 3.5 per cent of its stand-alone capital (H2), so stand-alone capital (Figure 3) overstates the median cost of a quote many times over. The endpoint instead returns the marginal capital against the maker's own book in one request.

Third, netting pays for books of a few dozen legs or more held under the portfolio managers (H3). The one standard-margin subaccount holds small books that net little; its operator keeps the large books under portfolio margin.

What does the edge cost?

13 events · 475 cell-event pairs · 91 446 rows from 84 919 fills (6 527 in two windows)

141 day clusters · 322 day × underlying effects

panel a: ▼ maker sells (short) · ▲ maker buys (long) · colour = manager

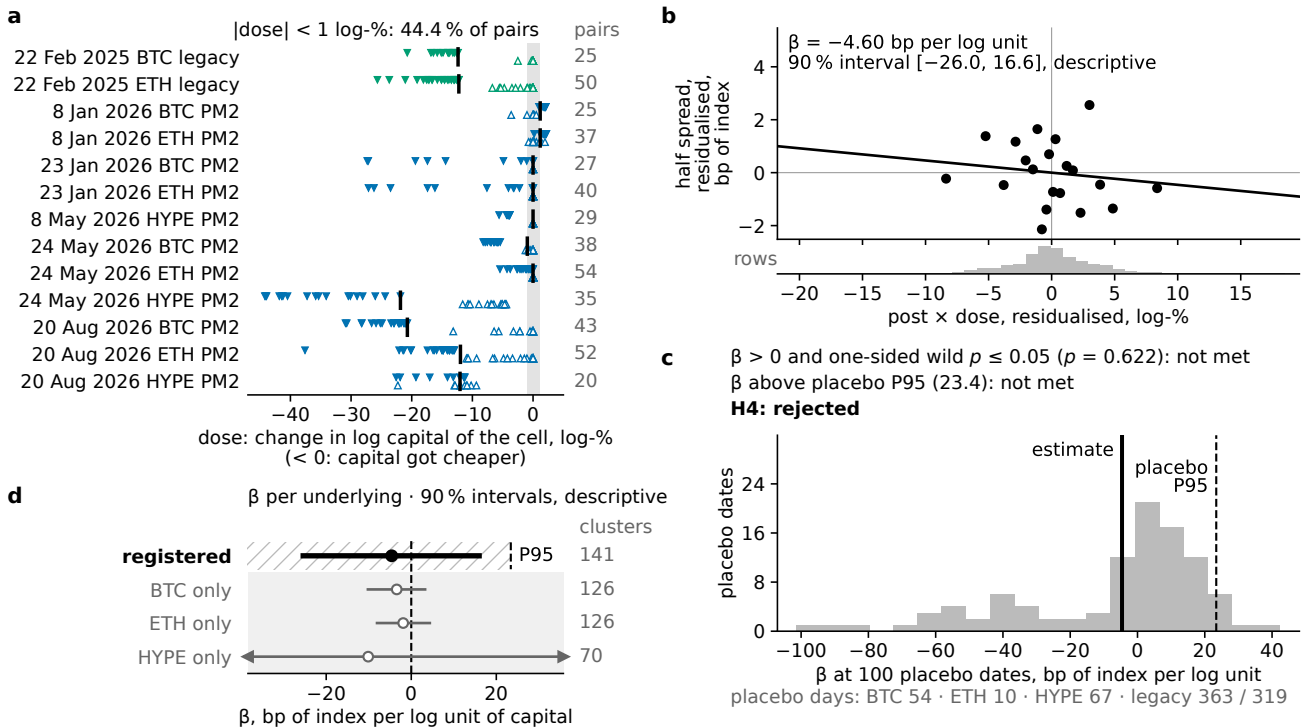


Figure 8: The price of capital (H4). Panel a shows the dose, the change in log capital that a parameter change makes to a cell, for each of the 475 cell-event pairs of the 13 events in the H4 panel, sells above and buys below each line, with the median as a bar. Panel b is the half spread against the post-event dose after removing the cell by event and the day by underlying effects, in 20 bins of equal size, above a grey histogram of that dose over the rows of the H4 panel, one per fill and event; the line shows the estimate β , whose interval is descriptive because the registered p comes from restricted residuals. That day-cluster interval is too narrow, as the placebo dates show; calibrated on them, the 90 per cent range runs from -41.6 to 29.7 (Section 4.4). Panel c places β among the estimates at 100 placebo dates and lists both registered criteria. Panel d gives β per underlying, for exploration; in the registered row, hatching marks the values of β at which the rule rejects H4 whatever its p , up to the 95th placebo percentile (P95) or zero, whichever is larger. Most doses are close to zero or negative, so β is identified from parameter changes that made capital cheaper. Doses on the axes are in log per cent, 100 times the change in log capital, so the line in panel b has the slope $\beta/100$ per log per cent. To read the slope: capital ten per cent cheaper is a dose of -0.105 , or -10.5 log per cent, which predicts a change in the half spread of -0.105β basis points of the index.

Fourth, parameters are a risk the maker does not control: a single change cut the PM2 capital of the BTC reference straddle by 22.9 per cent, and the half spread showed no detectable response (H4). Capital may have been slack, makers may have been bound by something else, or the test lacked power: for capital ten per cent cheaper, a narrowing by up to 39 per cent is not excluded (Section 4.4).

Several limits bound these readings. Capital is what the engine would require for the positions, not what makers deposited. Initial margin is assumed to bind. Margin for open orders, charged by the venue but not returned by the endpoint, lies outside the measure, as does every external bound on a book. Capital follows the on-chain engine, while the venue trades on the off-chain one. The net edge deducts a hedging cost, while capital is that of the unhedged contract. H2 rests on four PM2 subaccounts that trade ETH and HYPE only, H3 on books mostly larger than a standard-margin subaccount accepts, and H4 on 13 events that mostly made capital cheaper. Collateral other than USDC, held on most

maker-days, is left outside K . All numbers come from the pilot cut of 17 September 2026 and will be recomputed on the registered sample.

6. Conclusion

On Derive the capital behind a quote is not a private number. The rules that set it are public smart contracts with parameters on chain, so the capital of any book can be computed at any block, including books nobody held. Measured that way, capital does not reorder the maker's map in the registered sense (H1): the rank correlation is 0.903. Post hoc, much of that correlation comes from the sign of the edge, and the order changes more among profitable cells and at the top of the map. The median fill in a dominant PM2 book binds 3.5 per cent of its stand-alone capital (H2). Netting is worth a factor of 4.75 over standard margin on the books that the dominant subaccounts hold (H3), and most of these books are too large for a standard-margin

subaccount. The half spread showed no detectable response when parameter changes made capital cheaper (H4), but the test has little power. Four hypotheses were registered before any capital figure of the registered analysis existed; H1 and H4 are rejected, H2 and H3 are not. For a maker the capital that matters is that of the next fill in its own book, and the off-chain engine quotes it on request.

Data, code and pre-registration

Every input to this paper is public. The fills come from the venue's public trade tape as in Albiez (2026); feeds, parameters and maker holdings are read from the smart contracts on Derive Chain by historical `eth_call`; the present-day probes of the off-chain engine use its public endpoint (Derive, 2026a). The code is at github.com/gelatotrader/Derive-Option-Surface, together with its tests. It rebuilds the three managers from their smart contracts (Derive, 2026b), validates them against the chain, computes every capital figure, runs the inference and draws every figure. Before publication, an internal audit found errors in the code, the inference, the figures and the text. The file `docs/paper2/AUDIT.md` lists every finding and states how it was corrected or why it is still open; none changes a registered verdict.

The four hypotheses, their rejection rules and the inference were committed in `docs/paper2/PRAEREGISTRIRUNG.md` on 24 September 2026 at 22:33 UTC+2, in commit `c9e9162`, before any capital figure of the registered analysis had been computed (Burlig, 2018). The pre-registration is written in German, and its addenda are headed *Nachtrag*; an English translation made after the analysis is in `docs/paper2/PREREGISTRATION.md`, and the German original is binding. Seven dated addenda follow in the same file; the first four are of 25 September 2026 (UTC+2). Addendum 1 (`b5c905c`) fixes the exact on-chain book for H2, with zero cash, before the first capital figure of the registered form. Addendum 2 (`60896fe`) fixes the unit of fee and rebate, and Addendum 3 (`2416902`) the details of bootstrap and placebo, both before the first inference. Addendum 4 (`a02058a`) records the readings of H2 to H4. Addenda 1 to 3 were each committed before the step they govern. Addendum 4 was written after stand-alone capital, marginal costs, netting ratios, the doses and the H4 panel with its half spreads had been computed; it records readings already implemented, but precedes any test statistic. The validation fixtures committed just before Addendum 1 had already read the on-chain margins of individual maker subaccounts, cash included.

Probes preceded the registration: the probe books of Section 2, which set the threshold of H3; a map of edge per unit of SM capital over the BTC cells, from which the rank correlation that H1 tests could already be read for SM; and margin histories of at-the-money contracts under the three managers, which the registration does not list. The half spreads that enter H4 were known from the companion paper before registration; the registration fixes the doses, the events and the test.

All commit times are local times of the author's machine. Before publication the history was rewritten twice, first to remove account identifiers and then, for the whole repository, to make the authorship records uniform and the commit messages English. The registered texts are byte-identical to those of the original commits and the commit times are unchanged; `docs/paper2/HISTORY_REWRITE.md` maps the old commit hashes to the new ones. Addendum 5 of 27 September 2026 (`8c066a7`) and Addendum 7 of 5 October 2026 (`7c01e85`), both written after all results, record the rewrites and change no rule, test statistic or result. Addendum 6 of 30 September 2026 (`6f2436e`), written after the results on the pilot cut and before the final data run, keeps the fee of a multi-leg RFQ package on the leg on which the venue books it, as the companion paper does, and adds the fee spread over the package as a sensitivity of H1, under which ρ is 0.904 (Figure 4, panel c). The pre-registration became public with the push of this history on 5 October 2026, after the results on the pilot cut, so no public timestamp precedes the results. The history shows the order of the commits, not what was computed locally between them.

Competing interest

I am building a market-making bot for the venue studied here and have been in contact with its team about obtaining order-level data for a follow-up paper. No data, funding or review was received from the venue for this paper, and every number in it comes from public sources. Readers should weigh the practitioner conclusions in Section 5 in that light.

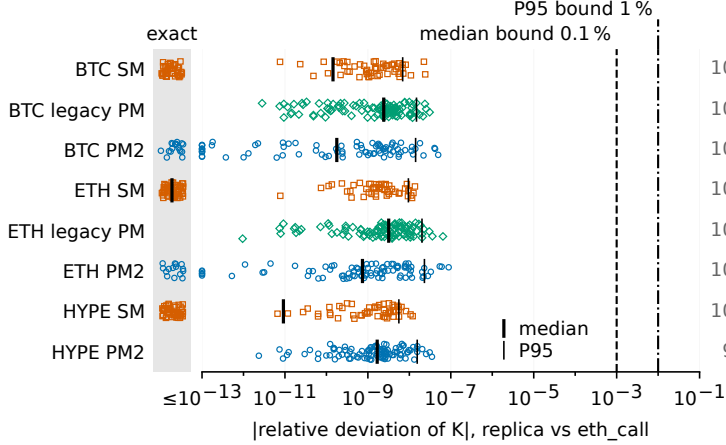
Use of generative tools

Large language models were used for code, for drafting and for checking the bibliography against publisher records. All results were produced by the code in the repository, and every published reference was verified against the publisher, RePEc or the DOI. The author is responsible for the content.

A. Validation of the replica

Figure 9 compares each offline replica with the deployed smart contracts, on 100 random blocks per underlying and manager (registered: at least 48) and on 20 maker-days. The 877 cases lie at blocks from 14 January 2024 to 17 September 2026, and one of them reverted on chain. All 32 cells of underlying, manager, kind and margin meet both registered thresholds. For single contracts under initial margin the median absolute relative deviation is 8.7×10^{-10} and the 95th percentile 1.4×10^{-8} ; over 799 contracts the largest is 8.9×10^{-8} . For 77 evaluations of the books of 20 maker-days under the available managers, with 2 to 245 legs, the median is 2.2×10^{-9} and the largest absolute deviation 0.050 USDC. In H3, 127 maker-days hold more legs than the largest validated book, up to 317; for these the replica is extrapolated in size but not in rule. For HYPE a single maker-day is validated. H2 also prices books larger than any validated one. No validation case falls in the settlement window before an

a single contracts



b opening books

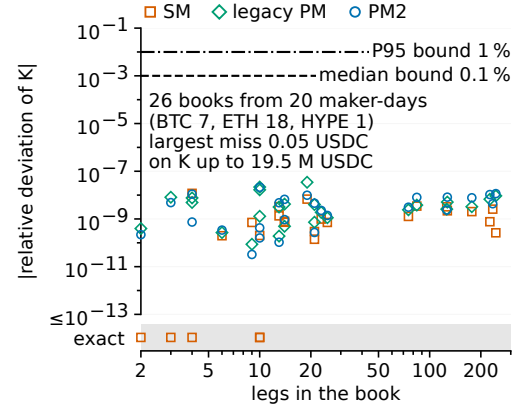


Figure 9: Does the replica match the chain? Absolute relative deviation of initial-margin capital between each offline replica and `eth_call` on the deployed smart contracts, for single contracts per underlying and manager (panel a) and for the opening books of 20 maker-days against their number of legs (panel b). Exact matches sit in the strip on the left of panel a and at the bottom of panel b; deviations below 10^{-13} sit on the edge of the axis. Thick ticks mark the median of a row, thin ticks its 95th percentile (P95). The lines are the registered bounds for the median (0.1 per cent) and the 95th percentile (1 per cent). One single HYPE contract under PM2 hit a reverting call and is left out.

expiry, when part of the forward is already fixed. In the PM2 window no fill uses a volatility feed older than 1 200 seconds or a forward feed older than 3 600 seconds, the feed ages the validation allows; 17 fills use a spot price older than the heartbeat of the spot feed (180 seconds) and stay in the sample.

B. Post hoc and exploratory results

Most of these analyses were added after the first results and are post hoc. The registration lists the maps under other managers, per underlying and per unit of time as exploratory, and the placebo rule over every timeline was computed with the results. None changes a verdict.

For H1, ranks shuffled within each sign group give a ρ of 0.734 on average, with 5th and 95th percentiles of 0.700 and 0.770. Among sells ρ is 0.990 and among buys 0.721. In the groups by the sign of the edge every replicate selects the cells by its own estimated edge; with the cells held fixed, the intervals would ignore that selection. Of the ten best cells per unit of capital, one is among the ten best per notional; of the best 20, five are. The percentile interval of the registered test is not centred on the estimate: 15 per cent of the draws lie at or above it, and a bias-corrected interval runs from 0.898 to 0.920. Maps in the PM2 window under the other managers or per underlying give ρ between 0.888 and 0.922; maps normalised by time to expiry or by the empirical holding period give 0.806 and 0.833 (Figure 4, panel c). Without RFQ fills ρ is 0.900 (0.889 to 0.913) over 153 cells. In the ten best cells per unit of PM2 capital, RFQ fills carry 89.6 per cent of the edge with 39.4 per cent of the fills.

For H2, the median runs from 0.0117 to 0.0899 over the parameter regimes R1 to R4, which the BTC changes of Figure 3 bound. By maker side the median is 0.075 (0.063 to 0.092) for sells and -0.137 (-0.202 to -0.075) for buys.

Weighted by stand-alone capital, as the sum of ΔK over the sum of stand-alone capital times size, all fills bind 0.166 of their stand-alone capital, sells 0.310, and buys release 0.397. Each leg of a multi-leg RFQ package is measured against the book before the whole package (Addendum 1), without the other legs. Collateral other than USDC, which the registration leaves outside K , is held on most maker-days. One PM2 subaccount holds ETH that its PM2 parameter set counts as offsetting risk, so counting that ETH would lower its PM2 capital.

For H4 the estimate is -3.42 for BTC, -1.90 for ETH and -10.12 for HYPE. It is -3.81 on sell cells alone and -5.13 with the dose weighted by the open-interest share of the changed manager. All of these estimates have a one-sided p above 0.5. On buy cells, whose doses are near zero, it is 128 with a p of 0.13. These p rest on the day-cluster standard error, which the placebo dates show to be too small. A wider reading of Addendum 4 keeps placebo dates at least 28 days from every parameter change of the underlying under any manager, including SM and the subaccounts' own PM2 parameter sets. It gives a 95th placebo percentile of 21.57, and 88 per cent of the placebo estimates are at least as large as β . The placebo panels hold one event more than the real panel, since the HYPE event of 8 January 2026 has no cell there. The placebo dates of ETH under PM2 come from 10 days between 16 July and 25 July 2025, so their windows overlap. With placebos only for events with cells and with disjoint windows per underlying, the 95th percentile is 26.02, 48 per cent of the placebo estimates are at least as large, and the placebo t has a standard deviation of 1.77. The dose is a mean of log ratios, and a few fills near zero capital move it. With the median log ratio as dose the estimate is -4.52 ; without fills whose log ratio exceeds one in absolute value, it is -4.08 .

References

- Ahn, J., 2025. Margin constraints and asset prices. *Review of Finance* 29, 141–168. doi:10.1093/rof/rfae039.
- Albiez, G., 2026. Who trades against the maker? Adverse selection with counterparty identity on an on-chain options order book. doi:10.2139/ssrn.7496818. SSRN working paper 7496818, University of Applied Sciences and Arts Northwestern Switzerland (FHNW), revised 25 September 2026.
- Artzner, P., Delbaen, F., Eber, J.M., Heath, D., 1999. Coherent measures of risk. *Mathematical Finance* 9, 203–228. doi:10.1111/1467-9965.00068.
- Avellaneda, M., Stoikov, S., 2008. High-frequency trading in a limit order book. *Quantitative Finance* 8, 217–224. doi:10.1080/14697680701381228.
- Brunnermeier, M.K., Pedersen, L.H., 2009. Market liquidity and funding liquidity. *Review of Financial Studies* 22, 2201–2238. doi:10.1093/rfs/hhn098.
- Burlig, F., 2018. Improving transparency in observational social science research: A pre-analysis plan approach. *Economics Letters* 168, 56–60. doi:10.1016/j.econlet.2018.03.036.
- Cameron, A.C., Gelbach, J.B., Miller, D.L., 2008. Bootstrap-based improvements for inference with clustered errors. *Review of Economics and Statistics* 90, 414–427. doi:10.1162/rest.90.3.414.
- Chen, H., Joslin, S., Ni, S.X., 2019. Demand for crash insurance, intermediary constraints, and risk premia in financial markets. *Review of Financial Studies* 32, 228–265. doi:10.1093/rfs/hhy004.
- Christoffersen, P., Goyenko, R., Jacobs, K., Karoui, M., 2018. Illiquidity premia in the equity options market. *Review of Financial Studies* 31, 811–851. doi:10.1093/rfs/hhx113.
- Comerton-Forde, C., Hendershott, T., Jones, C.M., Moulton, P.C., Seasholes, M.S., 2010. Time variation in liquidity: The role of market-maker inventories and revenues. *Journal of Finance* 65, 295–331. doi:10.1111/j.1540-6261.2009.01530.x.
- Cont, R., Kokholm, T., 2014. Central clearing of OTC derivatives: Bilateral vs multilateral netting. *Statistics & Risk Modeling* 31, 3–22. doi:10.1515/strm-2013-1161.
- Derive, 2026a. public/get_margin. Derive API reference. URL: https://docs.derive.xyz/api-reference/subaccounts/publicget_margin. Accessed 24 September 2026.
- Derive, 2026b. v2-core: Core smart contracts for the Lyra V2 protocol. GitHub repository derivexyz/v2-core. URL: <https://github.com/derivexyz/v2-core/tree/96796a6>. Commit 96796a6 of 16 February 2026, accessed 24 September 2026; Business Source License 1.1, licensor Lyra Foundation.
- Duffie, D., Zhu, H., 2011. Does a central clearing counterparty reduce counterparty risk? *Review of Asset Pricing Studies* 1, 74–95. doi:10.1093/rapstu/rar001.
- Figlewski, S., 1984. Margins and market integrity: Margin setting for stock index futures and options. *Journal of Futures Markets* 4, 385–416. doi:10.1002/fut.3990040307.
- Fournier, M., Jacobs, K., 2020. A tractable framework for option pricing with dynamic market maker inventory and wealth. *Journal of Financial and Quantitative Analysis* 55, 1117–1162. doi:10.1017/S0022109019000462.
- Gârleanu, N., Pedersen, L.H., 2011. Margin-based asset pricing and deviations from the law of one price. *Review of Financial Studies* 24, 1980–2022. doi:10.1093/rfs/hhr027.
- Gârleanu, N., Pedersen, L.H., Potoshman, A.M., 2009. Demand-based option pricing. *Review of Financial Studies* 22, 4259–4299. doi:10.1093/rfs/hhp005.
- Guéant, O., Lehalle, C.A., Fernandez-Tapia, J., 2013. Dealing with the inventory risk: A solution to the market making problem. *Mathematics and Financial Economics* 7, 477–507. doi:10.1007/s11579-012-0087-0.
- Ho, T., Stoll, H.R., 1981. Optimal dealer pricing under transactions and return uncertainty. *Journal of Financial Economics* 9, 47–73. doi:10.1016/0304-405X(81)90020-9.
- Jameson, M., Wilhelm, W., 1992. Market making in the options markets and the costs of discrete hedge rebalancing. *Journal of Finance* 47, 765–779. doi:10.1111/j.1540-6261.1992.tb04409.x.
- Kupiec, P.H., 1994. The performance of S&P 500 futures product margins under the SPAN margining system. *Journal of Futures Markets* 14, 789–811. doi:10.1002/fut.3990140704.
- Kupiec, P.H., White, A.P., 1996. Regulatory competition and the efficiency of alternative derivative product margining systems. *Journal of Futures Markets* 16, 943–968. doi:10.1002/(SICI)1096-9934(199612)16:8<943::AID-FUT6>3.0.CO;2-M.
- MacKinnon, J.G., Webb, M.D., 2017. Wild bootstrap inference for wildly different cluster sizes. *Journal of Applied Econometrics* 32, 233–254. doi:10.1002/jae.2508.
- Muravyev, D., 2016. Order flow and expected option returns. *Journal of Finance* 71, 673–708. doi:10.1111/jofi.12380.
- Qin, K., Zhou, L., Gamito, P., Jovanovic, P., Gervais, A., 2021. An empirical study of DeFi liquidations: Incentives, risks, and instabilities, in: *Proceedings of the 21st ACM Internet Measurement Conference (IMC '21)*, ACM. pp. 336–350. doi:10.1145/3487552.3487811.
- Roodman, D., Nielsen, M.Ø., MacKinnon, J.G., Webb, M.D., 2019. Fast and wild: Bootstrap inference in Stata using boottest. *Stata Journal* 19, 4–60. doi:10.1177/1536867X19830877.
- Santa-Clara, P., Saretto, A., 2009. Option strategies: Good deals and margin calls. *Journal of Financial Markets* 12, 391–417. doi:10.1016/j.finmar.2009.01.002.
- Soska, K., Dong, J.D., Khodaverdian, A., Zetlin-Jones, A., Routledge, B., Christin, N., 2021. Towards understanding cryptocurrency derivatives: A case study of BitMEX, in: *Proceedings of the Web Conference 2021 (WWW '21)*, ACM. pp. 45–57. doi:10.1145/3442381.3450059.
- Stoikov, S., Sağlam, M., 2009. Option market making under inventory risk. *Review of Derivatives Research* 12, 55–79. doi:10.1007/s11147-009-9036-3.